FinTechathon 2026 · International Track · Topic E | SME Finance Copilot

Vantora AI: An SME Finance Copilot for Cash Flow Recovery

AI as a Financial Participant



Team Vantora AI

Cash can cover the bill and still run out on payroll day

Receivables and obligations often do not land on the same day.

A treasurer usually sees the gap only when the bank balance turns negative. By then, three to five business days are already gone.

Every realistic recovery lever needs that time: an early-payment discount, a supplier deferral or a bridge draw.

Synthetic demonstration, day 14

Opening cash	$52,000
Safety buffer	$25,000
Payroll run, day 14	$68,000
Tooling payable, day 14	$35,000
P10 cash, day 14	−$49,700
Recovery needed to restore buffer	**$74,700**

Smart Jeans Manufacturing Ltd. is a fictional demo company. Source: Technical Documentation, section 3.1.5.

The agent proposes and drafts; the owner approves every commitment

What the agent does

- Reads company records, read-only and scoped to one company (Tier 0)
- Runs forecasts and recovery plans through the deterministic engine (Tier 0)
- Drafts offers, deferral requests and plan summaries, with no side effect (Tier 1)
- Prepares one exact action for owner approval (Tier 2)
- Explains results the engine has already computed

What it does not do

- Compute, estimate or invent a financial figure
- Execute any financial action without a single-use owner approval
- Move real money, or connect to a bank, payment rail or mail transport
- Change supplier bank details, disable audit logging or export records (Tier 3, prohibited)
- Act on its own: it runs from a user request and has no background scheduler

Topic E: three required capabilities, each mapped to a built module

Topic E requirement	Implemented feature	Module	Status
Financing-option matching	Three ordered screens (creditworthiness, funding speed, collateral), then ranking by effective APR	financing_ranker.py	Built, sandbox
Cash-flow management with alerts	30-day Monte Carlo forecast: first shortfall day, P10 buffer gap, decision day; Radar tab on the dashboard	forecaster.py, Radar tab	Forecast built; alerts TBD
Business financial analysis	Ledger-derived state (accounts, receivables, payables, facilities) and recovery optimiser with two counterfactual baselines	company-state.ts, optimizer.py	Built, sandbox

Built means described in the technical document with a named module. Not independently re-run for this deck. Real bank integration is not built.

Five planes; only the quantitative plane originates figures

Dashboard
Next.js 16 · Radar, Optimizer, Action Center, Ledger

→

AI orchestration
Function calls · 10 gated tools · bounded tool loop

→

Security kernel
Injection screen, tier check, attestation, approval, log

→

Quantitative plane
FastAPI · NumPy, SciPy, PuLP with HiGHS · sole source of figures

⇢

Persistence
SQLite for identity and approvals · PostgreSQL ledger mirror · SHA-256 chain

Company isolation

The company comes from the user's membership rows, never from the request. Roles are re-read from the database on every request. Cross-company access returns 403.

Control path

Every AI tool call passes the security kernel before any effect. Figures come only from the quantitative plane. Allow and deny decisions are both logged.

One task in six steps: the owner approves the exact action

1

Observe

Read ledger and financing state. Gaps appear as typed coverage notes.

2

Choose tools

The model picks from ten gated tools inside a bounded tool loop.

3

Compute

The engine runs forecast, screens and optimizer. The model computes nothing.

4

Draft

Plan summary or offer drafted. No side effect (Tier 1).

5

Approve

Owner checks exact vendor, amount and fee, types the key figure, and issues a single-use approval.

6

Record

Gateway executes atomically, consumes the nonce, and appends a SHA-256 receipt.

Every allow or deny decision is appended to the audit ledger before control returns.

Four deterministic algorithms; the model never sets a number

Forecast

1,000 seeded Monte Carlo runs, 30 days. Arrival = due date plus a truncated-normal delay, with a log-normal dispute tail.

Option ranking

Three ordered screens run first. Effective APR = (cost ÷ amount) × 365 ÷ 30 days. Example: 14.0% quoted plus 2.0% fee gives 38.3%.

Numerical verification

Each money figure in model text must match engine output within max($1, 0.5%); rates within 0.05 points. Signs are checked.

Recovery optimizer (MILP)

HiGHS solves a minimum-cost plan that restores the P10 buffer. Baselines: naive borrowing and do nothing.

Currency:
each company record carries a currency code, and approval payloads use integer cents. The sources describe no exchange-rate handling. **TBD.**

Simulated results from the canonical demo; sandbox logs still to be captured

Simulated, from the technical document

- Forecast: P10 cash on day 14 is −$49,700; shortfall in 100.0% of 1,000 trajectories
- Decision day 11 against a first shortfall on day 14: a three-day margin
- Heritage SBA (8.0% quoted) is rejected at Screen 2, because it cannot fund in time
- Recovery plan closes the $74,700 gap for $1,368.88 total cost
- Never more expensive than greedy over 216 scenarios; win count not stated: TBD

Sandbox operation log

TO BE CAPTURED

One recorded run of approval-to-execution on the simulated account: timestamps, token outcome and hash-chain entries.

Tests reported in the technical document: 220 automated (186 TypeScript, 34 Python). Self-reported; to be re-run before submission.

Security: a deterministic gate sits between the model and any effect

Permission tiers

Tier 0 · AUTO_READ	Read-only, runs unattended
Tier 1 · DRAFT_ONLY	Text only, no side effect
Tier 2 · HITL_APPROVAL	Runs only with a single-use owner approval
Tier 3 · PROHIBITED	Never runs for any caller: wire funds, change bank details, disable audit log, export records

Unregistered operations default to Tier 2.

Approval gate

HMAC-SHA256 token bound to the exact action, plan version and amount. Five-minute expiry, single-use nonce.

Data isolation

Company and role come from the database on every request, never from the client. Cross-company access returns 403.

Audit and sandbox

Each allow or deny joins a SHA-256 hash chain; payloads kept as digests. Simulated account only, no live payment rail.

Self-assessment probes, reported and not yet re-run: role forgery 403 · token replay 409 · amount tamper 422 · sign inversion 500 · invoice jailbreak isolated.

Known risks: what we have not solved yet

Risk	Severity	Current control	Open item
Password-only sign-in (R-01)	Medium	Hashed passwords, short sessions	Phishing can take a session; MFA planned
Multi-turn jailbreak (R-02)	Medium	Delimiters, strict JSON schema	Advanced jailbreaks may distort advice; classifier planned
Pattern-based injection screen	Medium	12 detectors; tier escalation	Novel phrasing can evade; not a fund control alone
Single ledger database (R-04)	Low	Hash chain, digests only	A DB superadmin could rewrite; root anchoring planned
Sandbox is not a live rail (R-05)	Low, by design	Atomic sandbox transactions	ACH and clearing delays not modelled
AI drafts not company-scoped	Medium	Owner review, Tier 2 approval	Open: wrong company name in draft text

The owner approves an exact action, not a chat message

What is new in this design

1. **The model never computes.** Figures come from the solver and are checked before a person sees them.
2. **Approval signs one action.** The owner types the exact figure; the approval is single-use and expires in five minutes.
3. **Visible gaps.** Missing or unsupported data appears as typed notes, never silently dropped.
4. **Counterfactual proof.** Each plan is priced against naive borrowing and doing nothing.

Owner interaction

1 · Review the exact vendor, amount and fee in the Action Center.

2 · Type the figure that matters; the server re-derives the plan first.

3 · Token issued; execution re-checks and consumes it in one transaction.

4 · A changed stress dial or counterparty response revokes outstanding approvals.

5 · Optional: delegate exactly the approved action; the model never sees the token.

Team Vantora AI

Taqi Ismail Mohammad

Team Leader · Backend Developer · AI/ML Expert

Imtiaj Bin Aoual

Full Stack Developer · AI/ML Expert · FinTech Expert

Sahib Abbas Bahar Chowdhury

Frontend Developer · AI Enthusiast · Product and Branding Specialist

In this prototype, the model explains and drafts, the engine computes, and the owner approves.

A sandbox prototype on a simulated account: no real money, no bank integration, no regulatory approval.

Source repository: github.com/taqiismail10/vantora-ai